

Improving Solubility and Reducing Aggregation of YpfH esterase by rational design of proteins

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Esterase YpfH

```
>NP_416968.2 esterase YpfH [Escherichia coli str. K-12 substr. MG1655]  
MKHDHFVVQSPDKPAQQLLLLFHGVGDNPVAMGEIGNWFAPLFPDALVVSVG  
GAEPSGNPAGRQWFSVQGITEDNRQARVDAIMPTFIETVRYWQKQSGVGAN  
ATALIGFSQGAIMVLESIKAEPGLASRVIAFNGRYASLPETASTATTIHLIHGGEDP  
VIDLAHAVAQAQALISAGGDVTLDIVEDLGHAIDNRSMQFALDHLRYTIPKH YFD  
EALSGGKPGDDDDVIEMM
```

BLASTp

We selected 5 sequences from BLASTp of YpfH against the SwissProt database and 30 sequences from BLASTp against the CLUSTERnr database. Overall, the identity score of the selected 24 sequences varied between 25% to 60%

BLAST[®] » blastp suite » results for RID-TZZ3MUJD016

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Database ClusteredNR [See details](#)
Query ID IclQuery_5827064
Description NP_416968.2 esterase YpfH [Escherichia coli str. K-12 sub ...
Molecule type amino acid
Query Length 232
Other reports [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

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	Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	1572 member(s), 53 organism(s) enterobacteria	enterobacteria	esterase [Escherichia coli]	458	458	100%	2e-162	96.98%	231	WP_042009324.1
☒	9 member(s), 4 organism(s) enterobacteria	enterobacteria	MULTISPECIES: esterase [Citrobacter]	425	425	100%	2e-149	87.50%	232	WP_012131307.1
☒	55 member(s), 12 organism(s) enterobacteria	enterobacteria	esterase [Citrobacter sp. RHB25-C09]	425	425	100%	2e-149	88.36%	231	WP_181681304.1

MSA

We used Clustal Omega to do MSA over the collected sequences.

- This will be later used to avoid mutating conserved positions.



Uniprot

According to Uniprot, the enzyme's active site involves three residues at positions 111 (S), 159 (D), and 191 (H). We should avoid mutations at these sites.

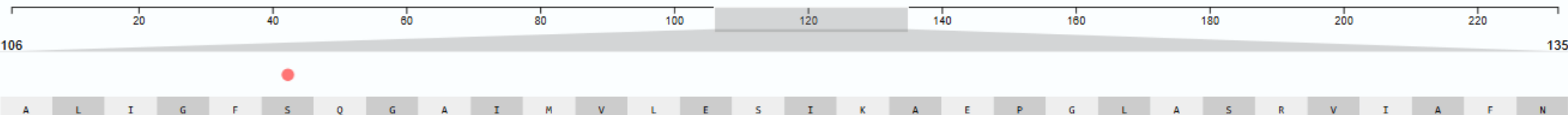
Functionⁱ

Displays esterase activity toward palmitoyl-CoA and pNP-butyrate. [1 Publication](#)

Features

Showing features for active siteⁱ.

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106 20 40 60 80 100 120 140 160 180 200 220 135

A L I G F S Q G A I M V L E S I K A E P G L A S R V I A F N

±	TYPE	ID	POSITION(S)	DESCRIPTION
+	Active site		111	Charge relay system By Similarity
+	Active site		159	Charge relay system By Similarity
+	Active site		191	Charge relay system By Similarity

MSA

This is further confirmed by our MSA data: positions 111, 159, and 191 are fully conserved in all of the 31 homologs analyzed.

Other conserved positions include:

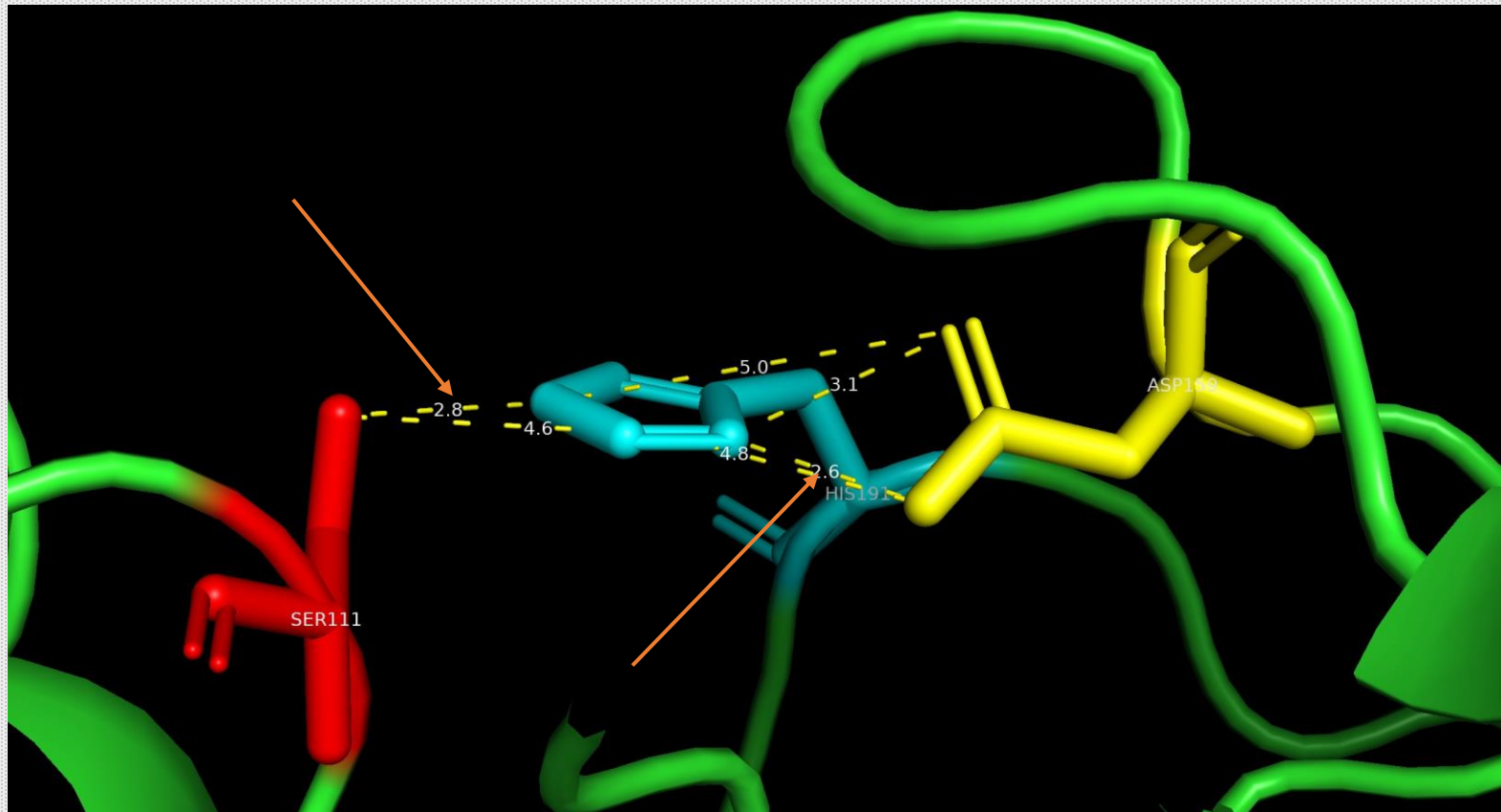
18L, 23H, 24G, 65W, 95Q, 105T, 109G, 110F, 112Q, 113G, 136G, 155H, 156G

VTIIIGFSQGTMI ^{GLY}	SELDOVV:	RLSHSIDGI
VTVIIGFSQGTII ^{GLY}	SELDOVIL	'SLAHSIDGI
VTVIIGFSQGTMI ^{GLY}	3ALDOVIL	'SLAHSIDGI
VTIIIGFSQGTMI ^{GLY}	SELDEVV	'SLAHSIDGI
VTIIIGFSQGTMI ^{GLY}	SELDEVV	'SLAHSIDGI
ITVLGFSQGAIMA ^{LE}	3DDDEVMI	IGVCHOPHP'
ITALGFSQGAIMA ^{LE}	3AADPVIL	'SVCHLHP,
ITLVGFSQGAIVS ^{LE}	3ADDAVIL	'GICHTLDP'
ITALIGFSQGSIMA ^{LE}	3SADPVIL	'GVCHLHA:
ITALFGFSQGAIMA ^{LE}	3SADPVIL	'GVCHLHA:
ITLVGFSQGSIMA ^{LE}	3DNDEMVI	'DLFHAIDEI
ITALIGFSQGAIMV ^{LE}	3GEDPVIL	'DLCHAIDNI
ITALIGFSQGGTMV ^{LE}	3DDQQVIL	'DLTHAIDHI
ITVLIGFSQGAIMS ^{LE}	3TSDPVIL	'KVAHGINP
ITLVGFSQGAIMA ^{LE}	3SDKVIL	'-VCHGISE,
ITVLVGFSGGSIMS ^{LE}	3DADGVIL	'AVCHGIDEI
ITTLIGFSQGSIMS ^{LE}	3EADGVIL	'GVCHGIDPI
ITTLIGFSQGSIMS ^{LE}	3ESDPVIL	'GVCHGIDQI
ITALLGFSQGATLS ^{LQ}	3ACDTLIL	'IDICHSVNQI
ITALLGFSQGAIMA ^{LE}	3KSDPVIL	'FICHOIHEI
ITALIGFSQGAIMS ^{LE}	3KADPVIL	'FICHTLHEI
ITLVGFSQGAIMALA	3KADTVMIL	'FICHEIHP
ITLVGFSQGAIMA ^{LE}	3KADAVIL	'FICHEIHP
ITALIGFSQGAIMA ^{LE}	3KODEVIL	'FVCHALVDI
ITALIGFSQGAIMA ^{LE}	3KLDEVIL	'FVCHALVDI
ITTLIGFSQGSIMS ^{LE}	3AADPVIL	'GLAHGIDAI
ITVLVGFSGGAIMS ^{LE}	3EODGVIL	'GLCHGIDDI
ITALIGFSQGAIMA ^{LE}	3KODPVIL	'DVCHGVPAL
ITLVGFSQGAIMA ^{LE}	3DDGVIL	'RLCHGVDKI
ITVLVGFSGGAMMV ^{LE}	3KVDIVIL	'FVCHALVDAI
ITVLVGFSGGASMV ^{LE}	3KTDEVIL	'LLCHGIDA
* : ** ** : *	* * .	: *
111S	159D	191H

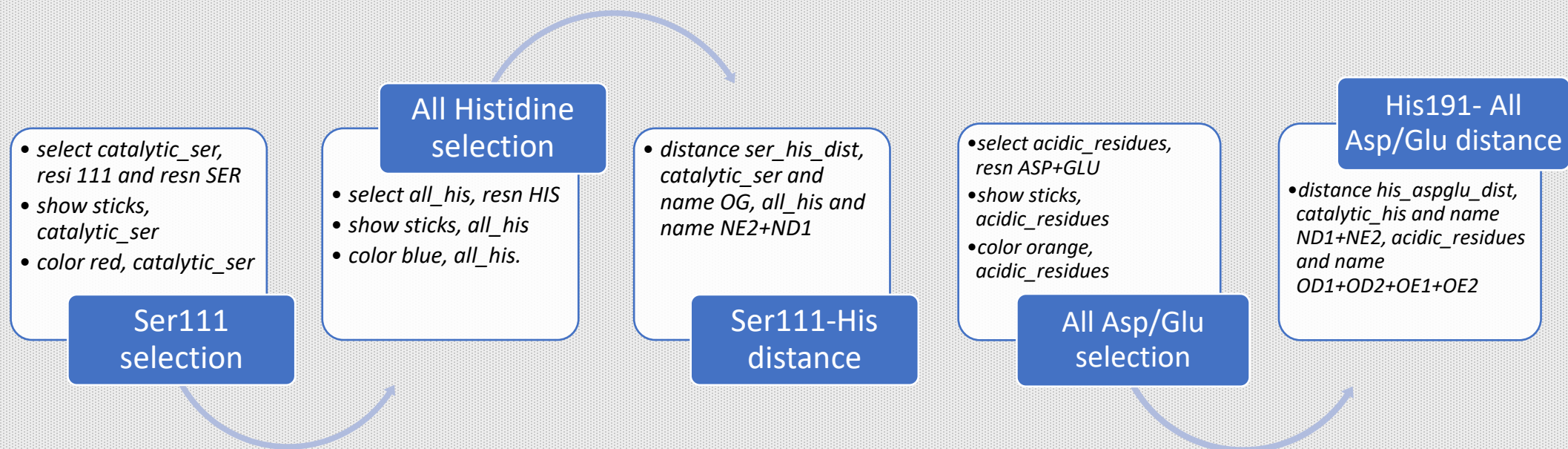
Catalytic triad in lipase/esterase

- Serine : nucleophile (attacks ester bond)
- Histidine : general base (activates serine)
- Aspartate or Glutamate : stabilizes histidine (charge relay system)

Within hydrogen bonding distance ($\sim 2.5\text{--}3.5\text{ \AA}$).



Validation of catalytic triad using PyMOL



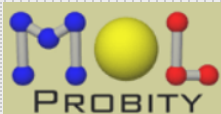
AlphaFold: 3D Modeling

We used AlphaFold3 to model the tertiary structure of the enzyme.



YfpH Esterase structure
determined via AF3

MolProbity: Model Validation



Analysis output: all-atom contacts and geometry for model.pdb



Duke Biochemistry
Duke University School of Medicine

Summary statistics

All-Atom Contacts	Clashscore, all atoms:	5.21		93 rd percentile* (N=1784, all resolutions)
	Clashscore is the number of serious steric overlaps (> 0.4 Å) per 1000 atoms.			
Protein Geometry	Poor rotamers	0	0.00%	Goal: <0.3%
	Favored rotamers	180	98.90%	Goal: >98%
	Ramachandran outliers	0	0.00%	Goal: <0.05%
	Ramachandran favored	226	98.26%	Goal: >98%
	Rama distribution Z-score	-0.42 ± 0.49		Goal: abs(Z score) < 2
	MolProbity score [^]	1.28		99 th percentile* (N=27675, 0Å - 99Å)
	Cβ deviations >0.25Å	0	0.00%	Goal: 0
	Bad bonds:	0 / 1793	0.00%	Goal: 0%
	Bad angles:	3 / 2441	0.12%	Goal: <0.1%
Peptide Omegas	Cis Prolines:	1 / 13	7.69%	Expected: ≤1 per chain, or ≤5%
Low-resolution Criteria	CaBLAM outliers	3	1.3%	Goal: <1.0%
	CA Geometry outliers	1	0.44%	Goal: <0.5%
Additional validations	Chiral volume outliers	0/271		
	Waters with clashes	0/0	0.00%	See UnDowser table for details

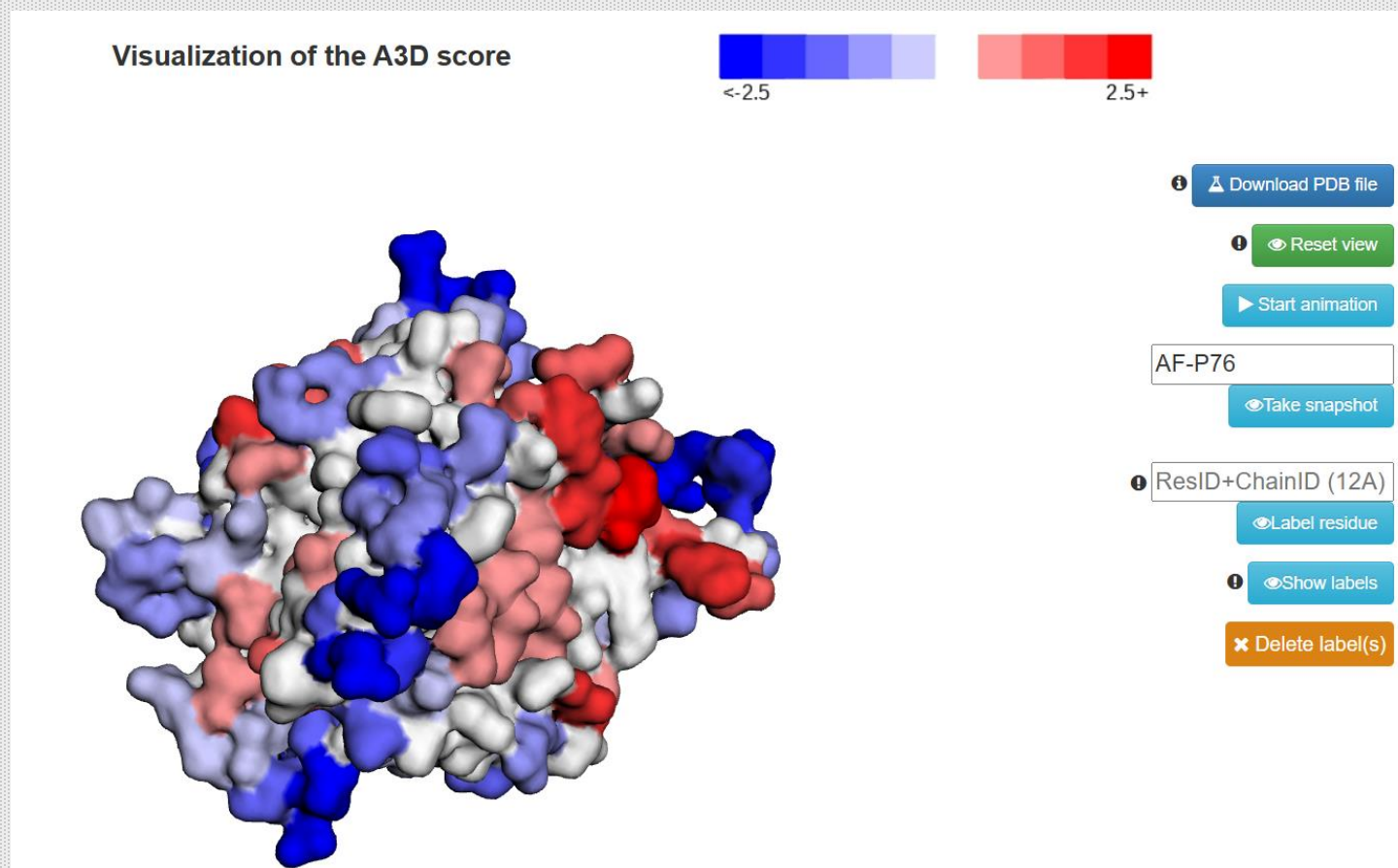
In the two column results, the left column gives the raw count, right column gives the percentage.

* 100th percentile is the best among structures of comparable resolution; 0th percentile is the worst. For clashscore the comparative set of structures was selected in 2004, for MolProbity score in 2006.

[^] MolProbity score combines the clashscore, rotamer, and Ramachandran evaluations into a single score, normalized to be on the same scale as X-ray resolution.

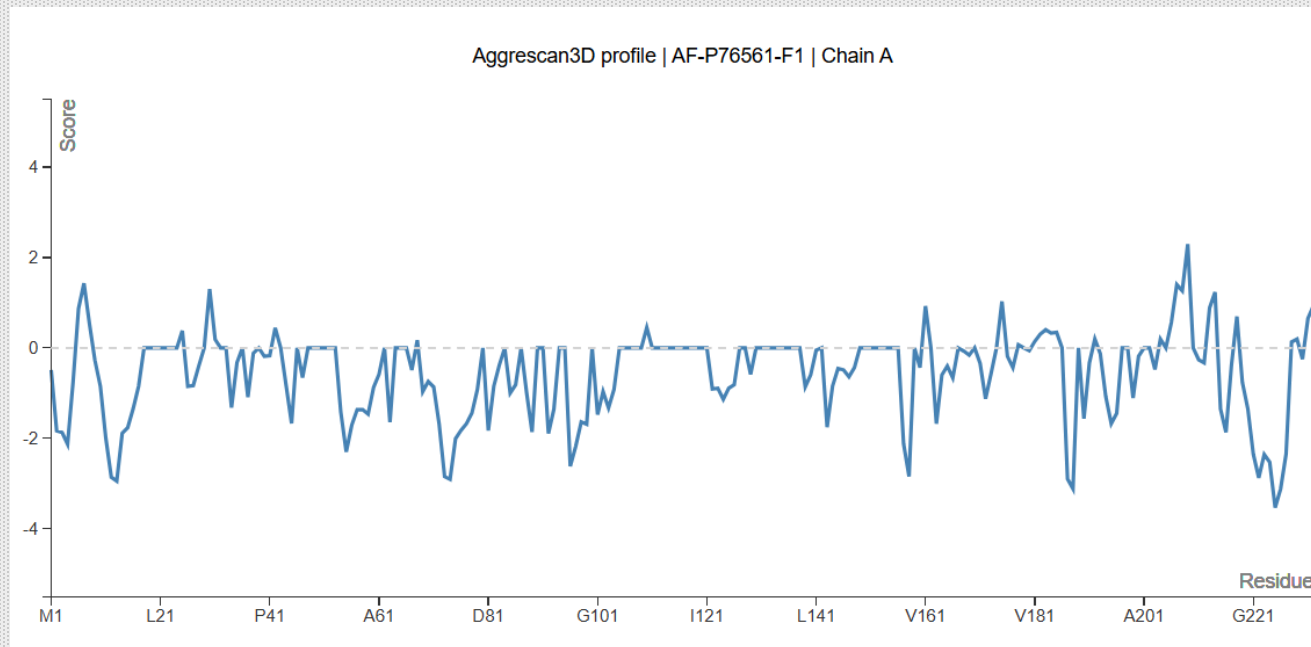
Key to table colors and cutoffs here: [🔗](#)

Aggrescan: prediction and engineering of protein solubility



Aggrescan: prediction and engineering of protein solubility

- A3D Average score: A normalized indicator of aggregation propensity/solubility of the protein structure. Allows comparing the solubility of different protein structures. It also allows assessing changes in solubility promoted by amino acid substitutions in a particular protein structure. The more negative the value the higher the normalized solubility.



Minimal score value	i -3.534
Maximal score value	i 2.2961
Average score	i -0.5513
Total score value	i -127.9095

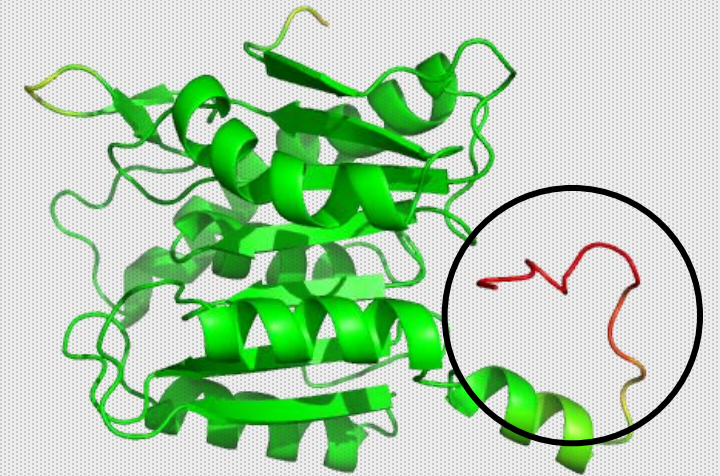
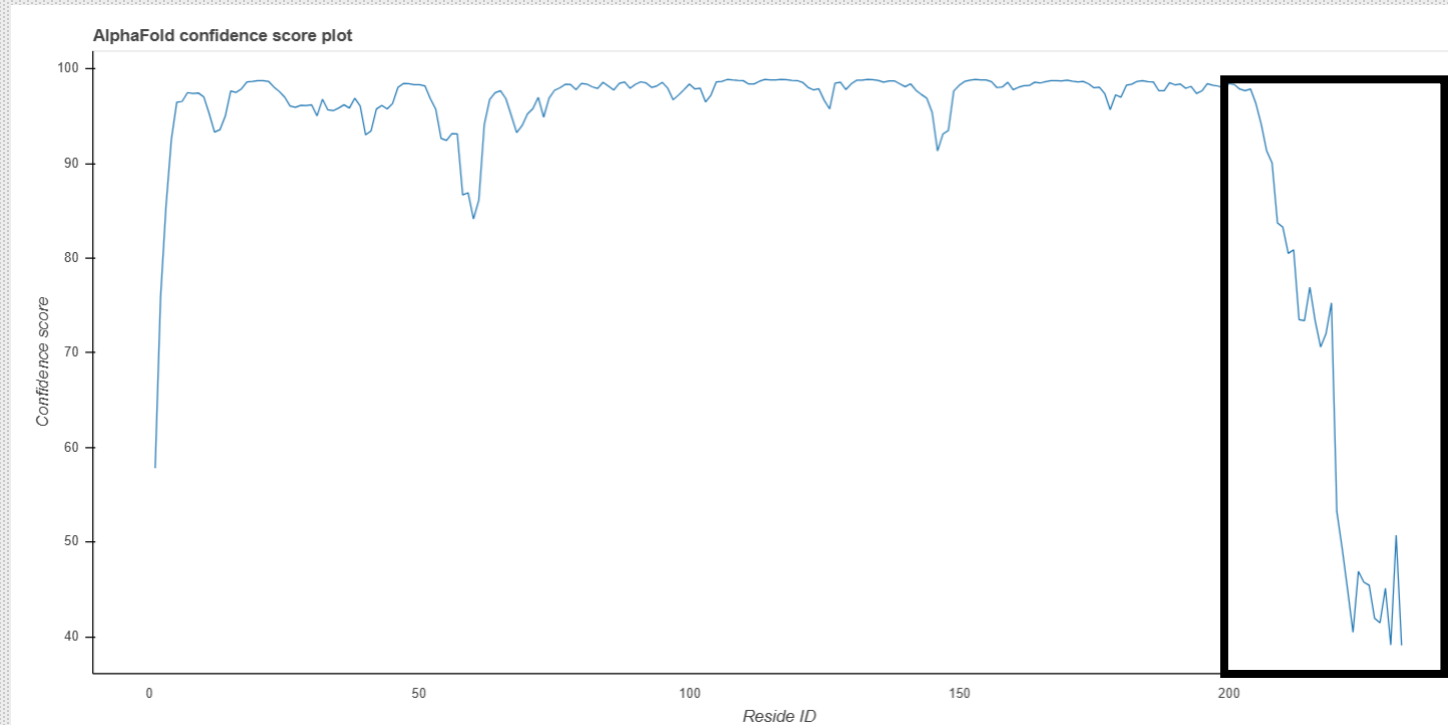
Aggrescan: prediction and engineering of protein solubility

- Aggrescan also uses FoldX to calculate $\Delta\Delta G$ between the mutated protein and the WT input. Stability calculations are crucial because we don't want a destabilized mutated protein!

Selected Chain(s)	A
Distance of aggregation	10 Å
Mutated residues ⓘ	YQ213A
Energy difference between WT (input) and mutated protein (by FoldX) ⓘ	0.241235 kcal/mol
Related PDB codes	None

Aggrescan: prediction and engineering of protein solubility

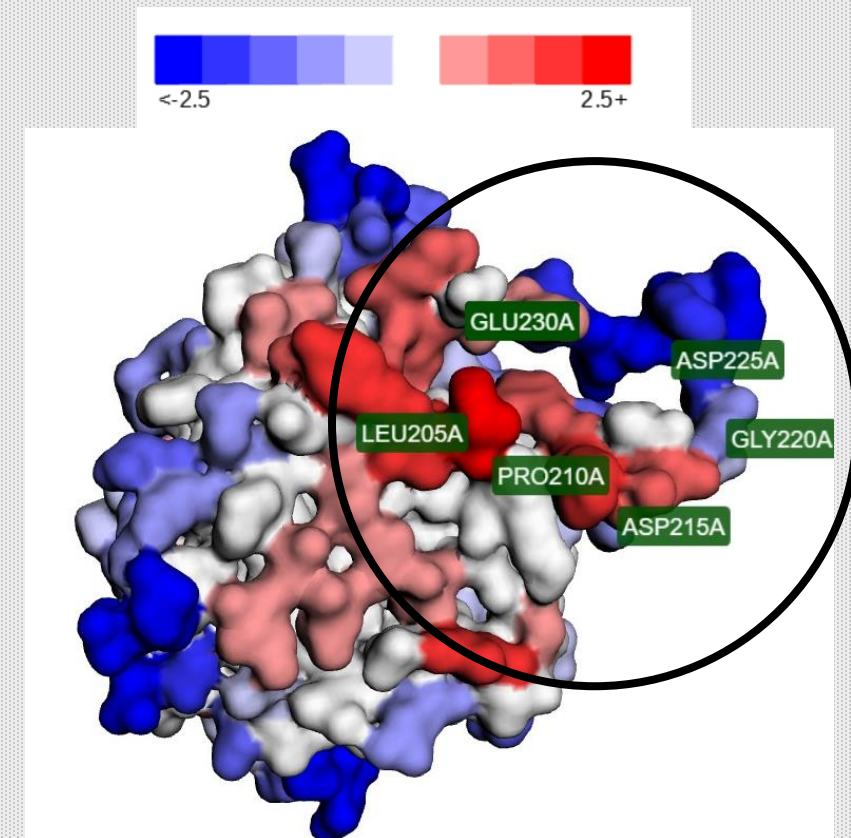
YpfH has a disordered C-terminal tail, as indicated by AF3 low confidence score (residues 200-232). Investigation of the 3D structure of the protein shows that this tail is exposed.



Aggrescan: prediction and engineering of protein solubility

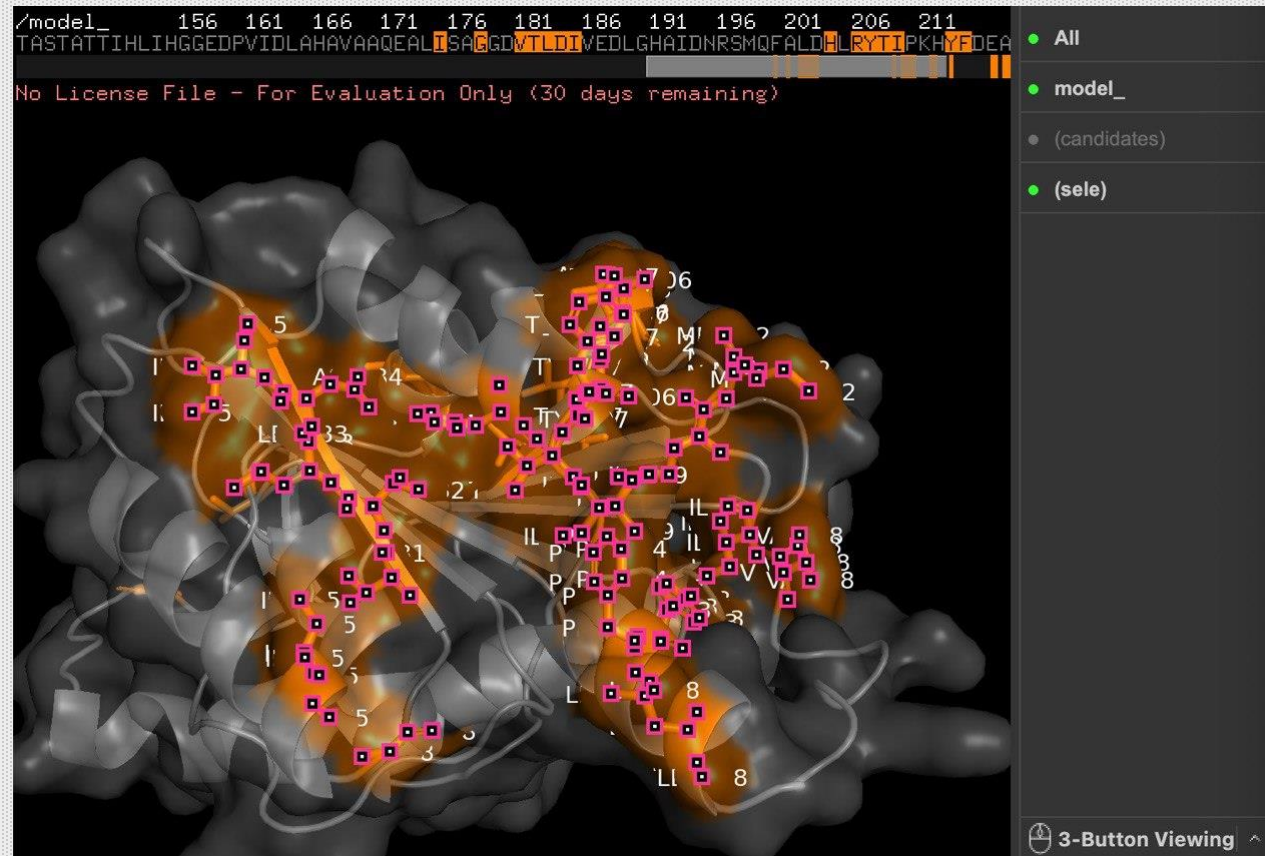
Moreover, this disordered tail has a high A3D score, and thus, many potential candidates. Using this rationale, we found 25 mutations (out of the total 38 mutations we found) in this region.

residue index	residue name	chain	Aggrescan3D score	AF Confidence
205	L	A	0.0000	96.37
206	R	A	0.5596	94.16
207	Y	A	1.3985	91.36
208	T	A	1.2650	90.09
209	I	A	2.2961	83.75
210	P	A	0.0000	83.29
211	K	A	-0.2605	80.52
212	H	A	-0.3288	80.91
213	Y	A	0.8829	73.52
214	F	A	1.2317	73.40
215	D	A	-1.3535	76.95
216	E	A	-1.8641	73.36

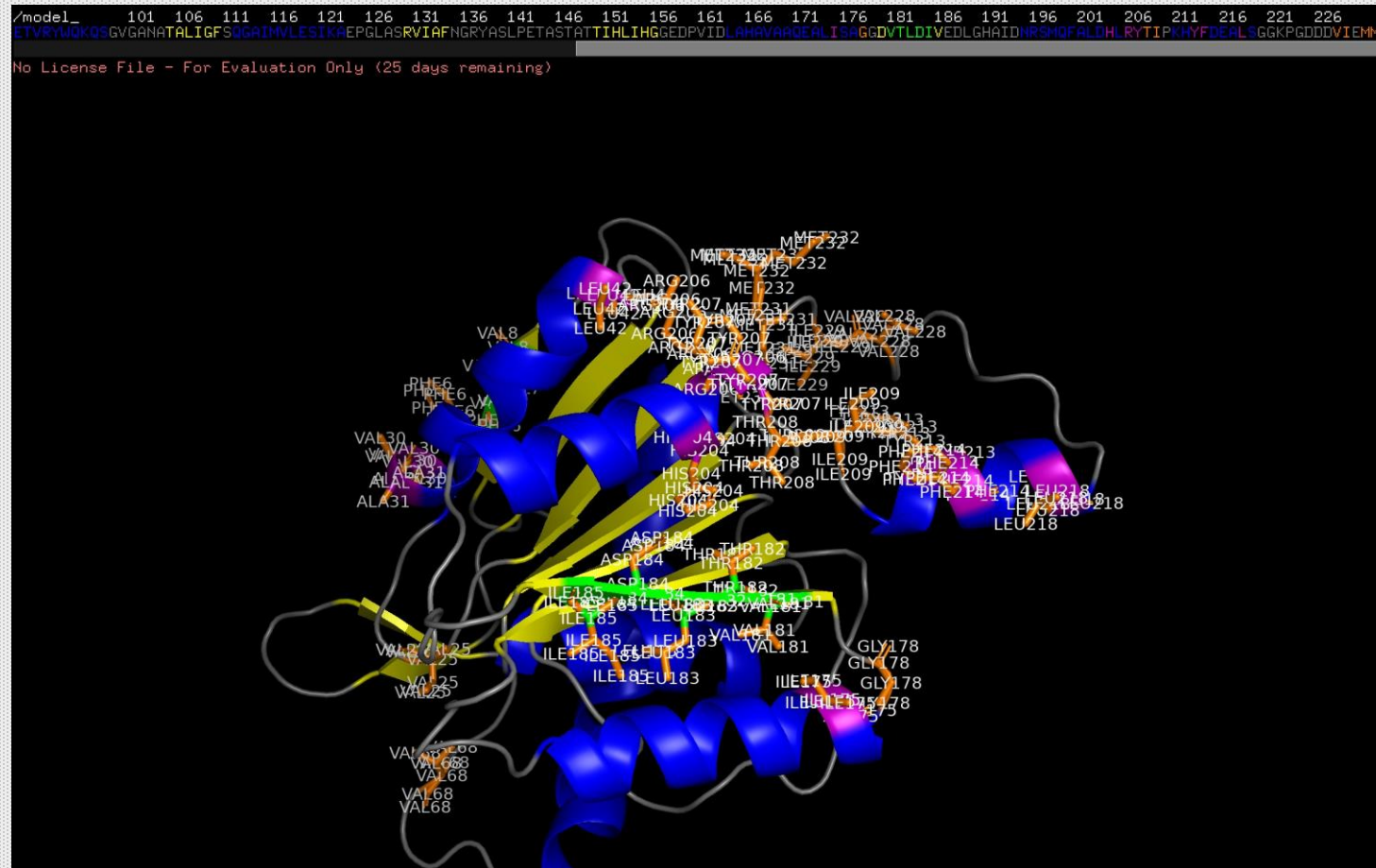


Visualization of aggregation-prone residues

- All A3D+ residues minus the ones near active site



Secondary structure assignment



Initial candidates

Determining factors:

- A3D score
- AF score
- Secondary structure

RESIDUE_ID	RESIDUE_NAME	CHAIN	A3D_SCORE	AF_SCORE	structure
6	F	A	0.8678	96.59	Beta sheet
7	V	A	1.4296	97.52	Beta sheet
8	V	A	0.5357	97.42	Beta sheet
25	V	A	0.3793	97.04	NA
30	V	A	1.3004	96.22	Alpha helice
31	A	A	0.1885	95.06	Alpha helice
42	L	A	0.4439	95.78	Alpha helice
68	V	A	0.17	93.28	NA
175	I	A	1.0282	98.03	Alpha helice
178	G	A	0.0711	95.71	NA
181	V	A	0.1425	98.3	Beta sheet
182	T	A	0.3006	98.41	Beta sheet
183	L	A	0.4007	98.71	Beta sheet
184	D	A	0.3291	98.79	Beta sheet
185	I	A	0.3437	98.67	Beta sheet
204	H	A	0.1957	97.91	Alpha helice
206	R	A	0.5596	94.16	Alpha helice
207	Y	A	1.3985	91.36	Alpha helice
208	T	A	1.265	90.09	NA
209	I	A	2.2961	83.75	NA
213	Y	A	0.8829	73.52	Alpha helice
214	F	A	1.2317	73.4	Alpha helice
218	L	A	0.6953	72.04	Alpha helice
228	V	A	0.131	41.46	NA
229	I	A	0.1937	45.12	NA
231	M	A	0.6531	50.72	NA
232	M	A	0.9324	39.08	NA

Raw results (sorted based on A3D average scores)

mutation	A3D_score	Δ A3D_score	$\Delta\Delta G$ (kcal/mol)
VR228A	-0.6046	-0.0533	-0.9943
IR175A	-0.5909	-0.0396	-0.09
LR218A	-0.5883	-0.037	-0.636501
FR214A	-0.5835	-0.0322	-0.034
IN175A	-0.5818	-0.0305	0.95
VE228A	-0.5818	-0.0305	-0.37
IQ175A	-0.5811	-0.0298	0.26
VE30A	-0.5782	-0.0269	-0.38
VN228A	-0.578	-0.0267	-0.44
YE207A	-0.576	-0.0247	0.44
LD218A	-0.5759	-0.0246	0.62
ID229A	-0.5745	-0.0232	0.75
IQ209A	-0.5729	-0.0216	0.56
IN209A	-0.5725	-0.0212	0.85
VK30A	-0.5723	-0.021	-0.46886
YR207A	-0.5717	-0.0204	-0.7996673
IE229A	-0.5703	-0.019	0.92
YE213A	-0.5699	-0.0186	0.87
LN218A	-0.5697	-0.0184	-0.034

mutation	A3D_score	Δ A3D_score	$\Delta\Delta G$ (kcal/mol)
VN30A	-0.5696	-0.0183	0.006
YN207A	-0.5679	-0.0166	0.17
IN229A	-0.5679	-0.0166	0.24
VE161A	-0.5668	-0.0155	0.35
YQ207A:	-0.5644	-0.0131	0.35
VN161A	-0.5635	-0.0122	0.54
IS175A	-0.5629	-0.0116	0.62
IQ229A	-0.5621	-0.0108	0.56
LQ218A	-0.5614	-0.0101	-0.18
VD68A	-0.5614	-0.0101	0.46
VR68A	-0.5586	-0.0073	-0.014
IR229A	-0.5571	-0.0058	0.168
MR232A	-0.5561	-0.0048	-0.5
IG229A	-0.5559	-0.0046	0.95
ME232A	-0.5559	-0.0046	0.19
MD232A	-0.5557	-0.0044	0.27
MK232A	-0.5545	-0.0032	-0.46
VE7A	-0.5541	-0.0028	0.73

Initial Filtering (16 mutations)

mutation	A3D_score	Δ A3D_score	$\Delta\Delta G$ (kcal/mol)
VR228A	-0.6046	-0.0533	-0.9943
IR175A	-0.5909	-0.0396	-0.09
LR218A	-0.5883	-0.037	-0.636501
FR214A	-0.5835	-0.0322	-0.034
IN175A	-0.5818	-0.0305	0.95
VE228A	-0.5818	-0.0305	-0.37
IQ175A	-0.5811	-0.0298	0.26
VE30A	-0.5782	-0.0269	-0.38
VN228A	-0.578	-0.0267	-0.44
YE207A	-0.576	-0.0247	0.44
LD218A	-0.5759	-0.0246	0.62
ID229A	-0.5745	-0.0232	0.75
IQ209A	-0.5729	-0.0216	0.56
IN209A	-0.5725	-0.0212	0.85
VK30A	-0.5723	-0.021	-0.46886
YR207A	-0.5717	-0.0204	-0.7996673
IE229A	-0.5703	-0.019	0.92
YE213A	-0.5699	-0.0186	0.87
LN218A	-0.5697	-0.0184	-0.034

mutation	A3D_score	Δ A3D_score	$\Delta\Delta G$ (kcal/mol)
VN30A	-0.5696	-0.0183	0.006
YN207A	-0.5679	-0.0166	0.17
IN229A	-0.5679	-0.0166	0.24
VE161A	-0.5668	-0.0155	0.35
YQ207A:	-0.5644	-0.0131	0.35
VN161A	-0.5635	-0.0122	0.54
IS175A	-0.5629	-0.0116	0.62
IQ229A	-0.5621	-0.0108	0.56
LQ218A	-0.5614	-0.0101	-0.18
VD68A	-0.5614	-0.0101	0.46
VR68A	-0.5586	-0.0073	-0.014
IR229A	-0.5571	-0.0058	0.168
MR232A	-0.5561	-0.0048	-0.5
IG229A	-0.5559	-0.0046	0.95
ME232A	-0.5559	-0.0046	0.19
MD232A	-0.5557	-0.0044	0.27
MK232A	-0.5545	-0.0032	-0.46
VE7A	-0.5541	-0.0028	0.73

CamSol Intrinsic Score (Criterion: $\Delta\text{score} > 0$)

mutation	A3D_score	$\Delta\text{A3D_score}$	$\Delta\Delta\text{G (kcal/mol)}$	$\Delta\text{CamSol_Intinsic_Score}$
VR228A	-0.6046	-0.0533	-0.9943	0.093765
IR175A	-0.5909	-0.0396	-0.09	0.005472
LR218A	-0.5883	-0.037	-0.636501	0.156629
FR214A	-0.5835	-0.0322	-0.034	0.057282
IN175A	-0.5818	-0.0305	0.95	0.000193
VE228A	-0.5818	-0.0305	-0.37	0.121432
IQ175A	-0.5811	-0.0298	0.26	0
VE30A	-0.5782	-0.0269	-0.38	0.058222
VN228A	-0.578	-0.0267	-0.44	0.080285
YE207A	-0.576	-0.0247	0.44	0.038931
LD218A	-0.5759	-0.0246	0.62	0.067796
ID229A	-0.5745	-0.0232	0.75	0.095812
IQ209A	-0.5729	-0.0216	0.56	0.036927
IN209A	-0.5725	-0.0212	0.85	0.036927
VK30A	-0.5723	-0.021	-0.46886	0.124821
YR207A	-0.5717	-0.0204	-0.7996673	0.047739

CamSol Structurally Corrected Score (Criterion: $\Delta\text{score} > 0$)

mutation	A3D_score	$\Delta\text{A3D_score}$	$\Delta\Delta\text{G}$ (kcal/mol)	$\Delta\text{CamSol_Intinsic_Score}$	$\Delta\text{CamSol_structurally_corrected}$
VR228A	-0.6046	-0.0533	-0.9943	0.093765	2.78
IR175A	-0.5909	-0.0396	-0.09	0.005472	1.71
LR218A	-0.5883	-0.037	-0.636501	0.156629	9.49
FR214A	-0.5835	-0.0322	-0.034	0.057282	3.12
IN175A	-0.5818	-0.0305	0.95	0.000193	-0.99X
VE228A	-0.5818	-0.0305	-0.37	0.121432	5.74
IQ175A	-0.5811	-0.0298	0.26	0	-0.22X
VE30A	-0.5782	-0.0269	-0.38	0.058222	2.36
VN228A	-0.578	-0.0267	-0.44	0.080285	2.8
YE207A	-0.576	-0.0247	0.44	0.038931	4.24
LD218A	-0.5759	-0.0246	0.62	0.067796	3.22
ID229A	-0.5745	-0.0232	0.75	0.095812	4.59
IQ209A	-0.5729	-0.0216	0.56	0.036927	-1.26X
IN209A	-0.5725	-0.0212	0.85	0.036927	-0.59X
VK30A	-0.5723	-0.021	-0.46886	0.124821	6.6
YR207A	-0.5717	-0.0204	-0.7996673	0.047739	1.69

Dynamut (Criterion: $\Delta\Delta G \leq -1.0$ kcal/mol)

mutation	A3D_score	$\Delta A3D_score$	$\Delta\Delta G$ (kcal/mol)	$\Delta CamSol_Intinsic_Score$	$\Delta CamSol_structurally_corrected$	Dynamut score
VR228A	-0.6046	-0.0533	-0.9943	0.093765	2.78	-0.293
IR175A	-0.5909	-0.0396	-0.09	0.005472	1.71	0.765
LR218A	-0.5883	-0.037	-0.636501	0.156629	9.49	-0.279
FR214A	-0.5835	-0.0322	-0.034	0.057282	3.12	-0.472
IN175A	-0.5818	-0.0305	0.95	0.000193	-0.99X	0.016
VE228A	-0.5818	-0.0305	-0.37	0.121432	5.74	-0.082
IQ175A	-0.5811	-0.0298	0.26	0	-0.22X	-0.407
VE30A	-0.5782	-0.0269	-0.38	0.058222	2.36	-0.088
VN228A	-0.578	-0.0267	-0.44	0.080285	2.8	-0.168
YE207A	-0.576	-0.0247	0.44	0.038931	4.24	0.422
LD218A	-0.5759	-0.0246	0.62	0.067796	3.22	-0.273
ID229A	-0.5745	-0.0232	0.75	0.095812	4.59	-0.212
IQ209A	-0.5729	-0.0216	0.56	0.036927	-1.26X	-0.051
IN209A	-0.5725	-0.0212	0.85	0.036927	-0.59X	-0.465
VK30A	-0.5723	-0.021	-0.46886	0.124821	6.6	-0.078
YR207A	-0.5717	-0.0204	-0.7996673	0.047739	1.69	-0.235

Electrostatic potential map of the protein surface

Step 1: Removing Solvent and Unnecessary Molecules:

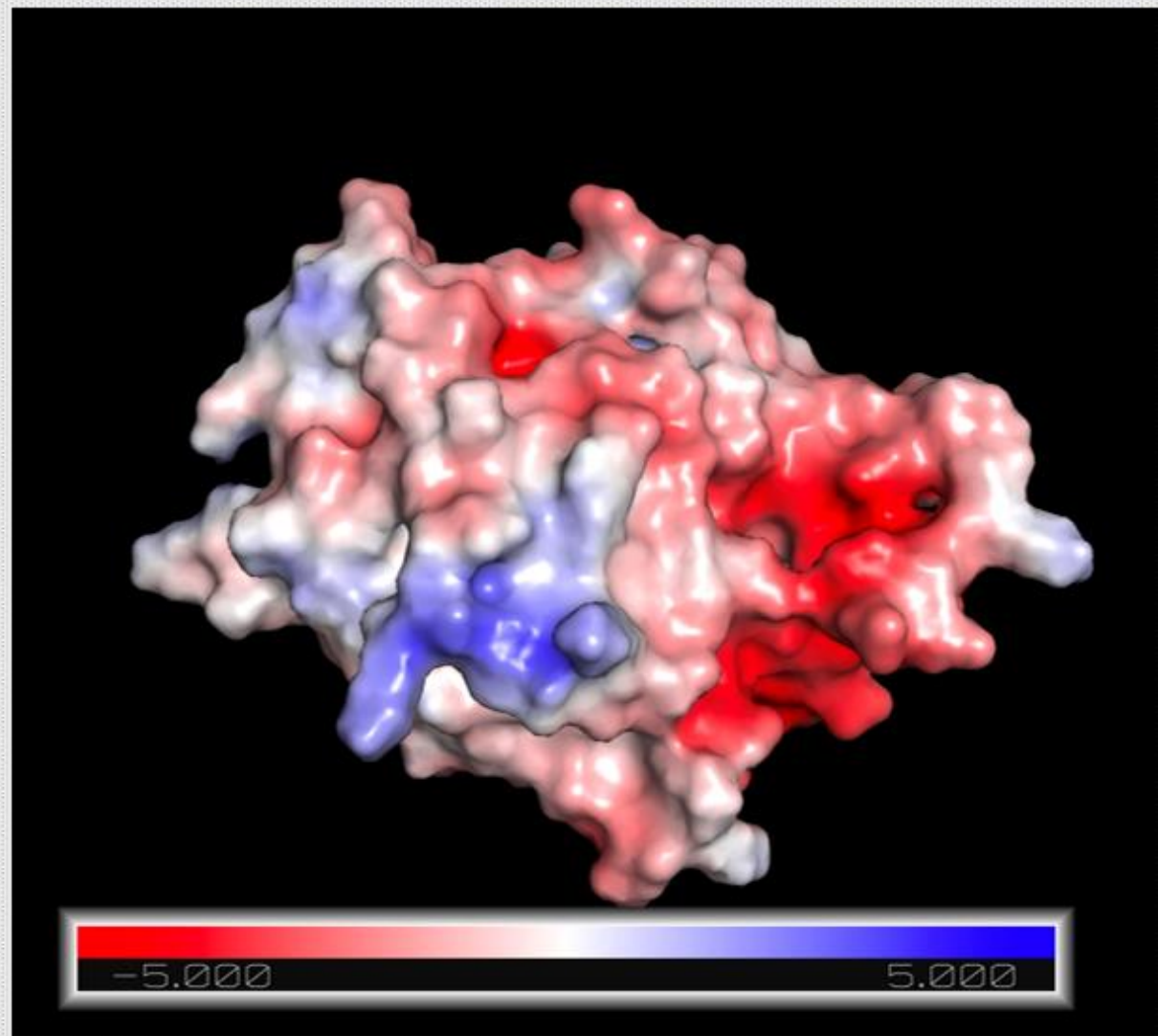
remove solvent

remove resn HOH

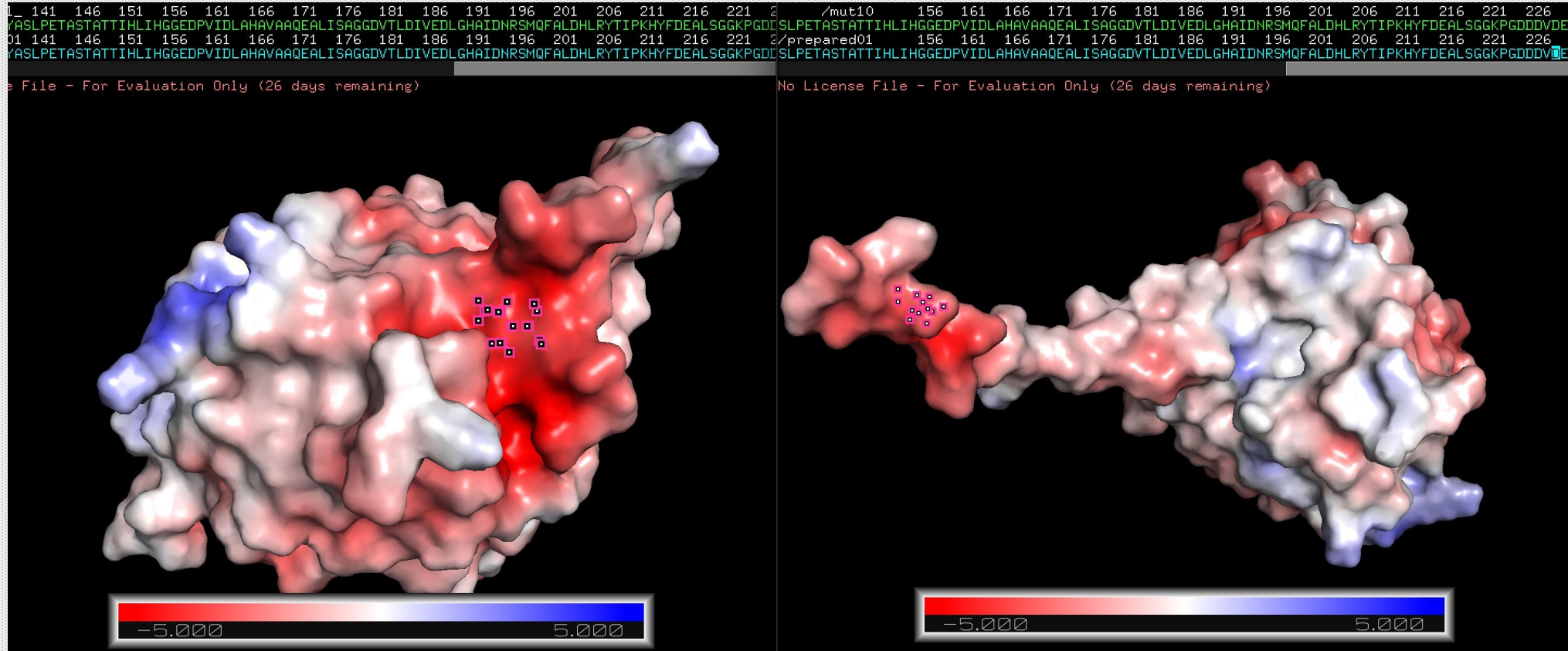
Step 2:

h_add

Step 3: Running APBS(Adaptive Poisson–Boltzmann Solver)



Removing some candidates through tertiary structure analysis



ID229 Mutation X

SASA (Solvent Accessible surface area)

```
PyMOL>get_area resi 175  
cmd.get_area: 112.660 Angstroms^2.
```

$\text{SASA}/\text{Max SASA} = \text{Exposure}$

Residue	Max ASA(A2)	Residue	Max ASA(A2)
A	106	E	194
L	164	H	184
I	169	N	157
V	142	Q	198
G	84	S	130
K	205	T	142
R	248	C	135
D	163	M	188
Y	222	F	197
W	227	P	136

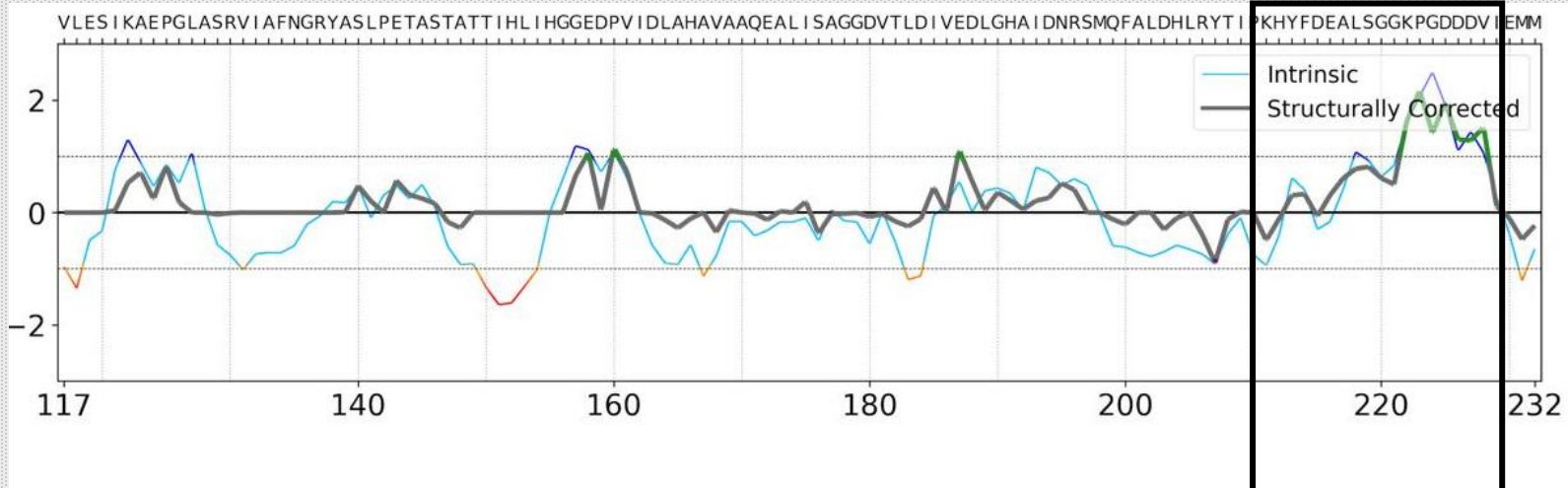
Final list

We selected 3 mutations based on their overall scores on our metrics for final analysis.

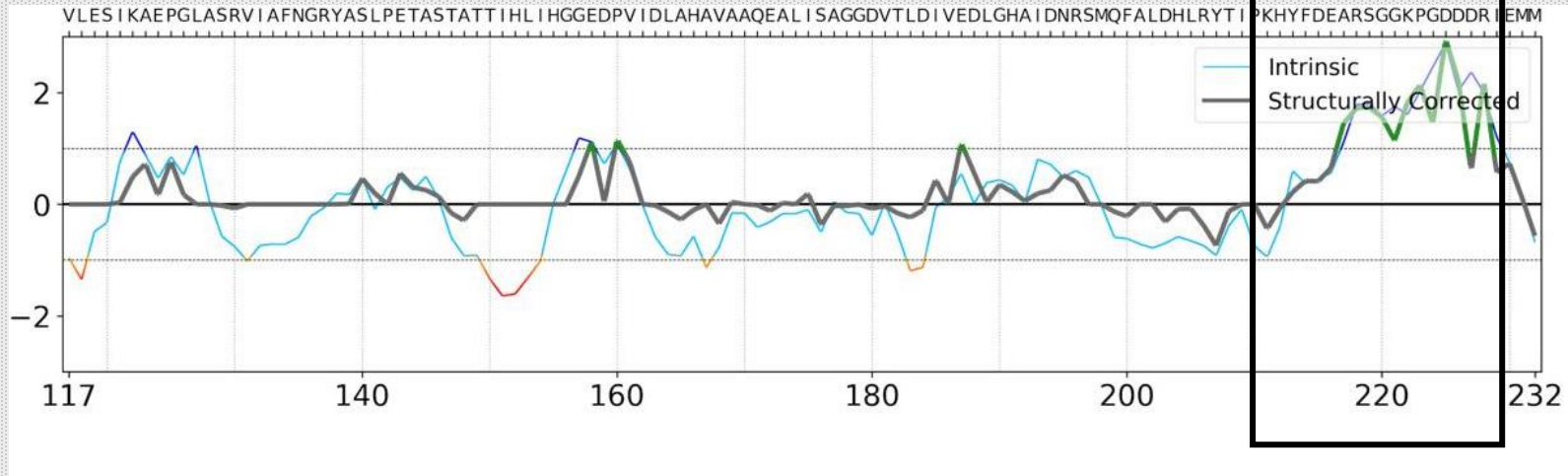
mutation	A3D_score	Δ A3D_score	$\Delta\Delta G$ (kcal/mol)	Δ CamSol_Intinsic_Score	Δ CamSol_structurally_corrected	Dynamut score
VR228A	-0.6046	-0.0533	-0.9943	0.093765	2.78	-0.293
IR175A	-0.5909	-0.0396	-0.09	0.005472	1.71	0.765
LR218A	-0.5883	-0.037	-0.636501	0.156629	9.49	-0.279
FR214A	-0.5835	-0.0322	-0.034	0.057282	3.12	-0.472
VE228A	-0.5818	-0.0305	-0.37	0.121432	5.74	-0.082
VE30A	-0.5782	-0.0269	-0.38	0.058222	2.36	-0.088
VN228A	-0.578	-0.0267	-0.44	0.080285	2.8	-0.168
YE207A	-0.576	-0.0247	0.44	0.038931	4.24	0.422
LD218A	-0.5759	-0.0246	0.62	0.067796	3.22	-0.273
VK30A	-0.5723	-0.021	-0.46886	0.124821	6.6	-0.078
YR207A	-0.5717	-0.0204	-0.7996673	0.047739	1.69	-0.235

LR218 CamSol Profile

WT

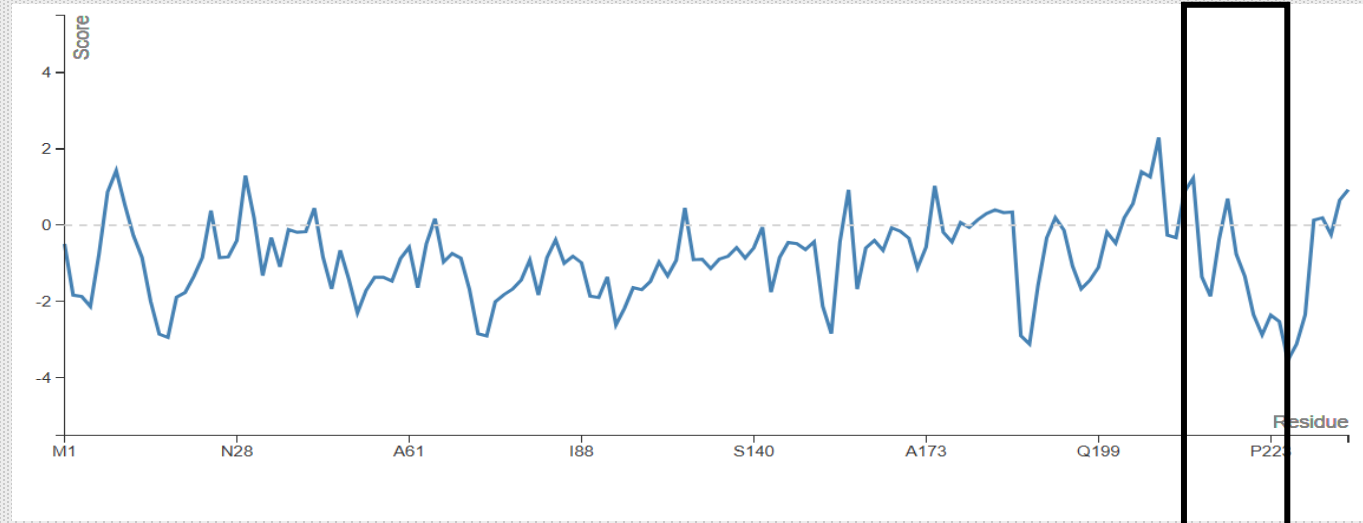


Mutant

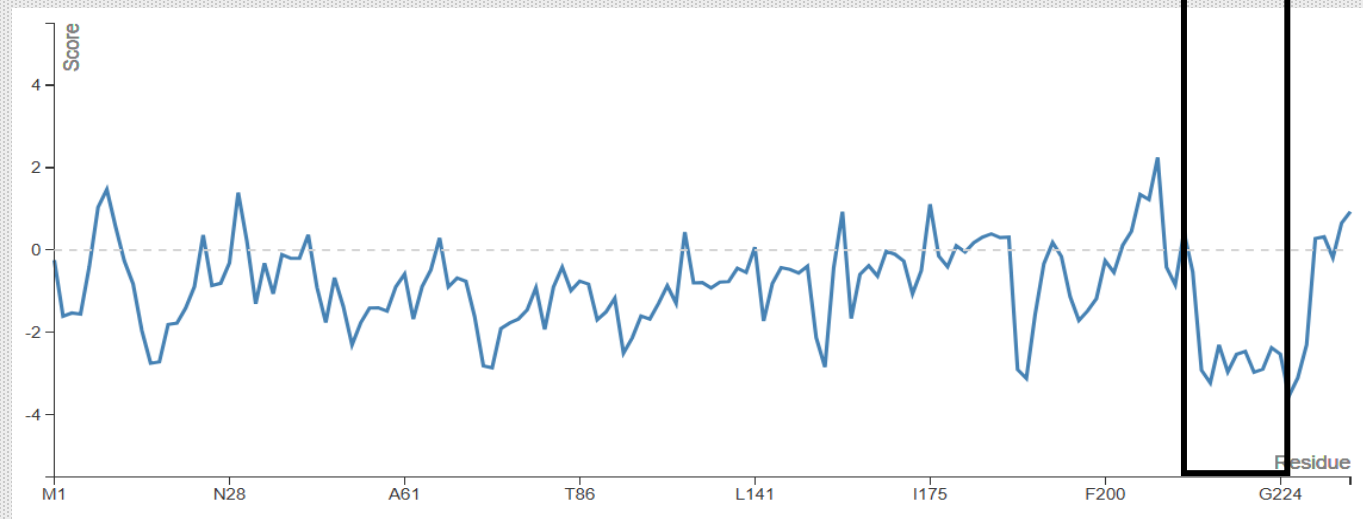


LR218 Aggrescan Profile

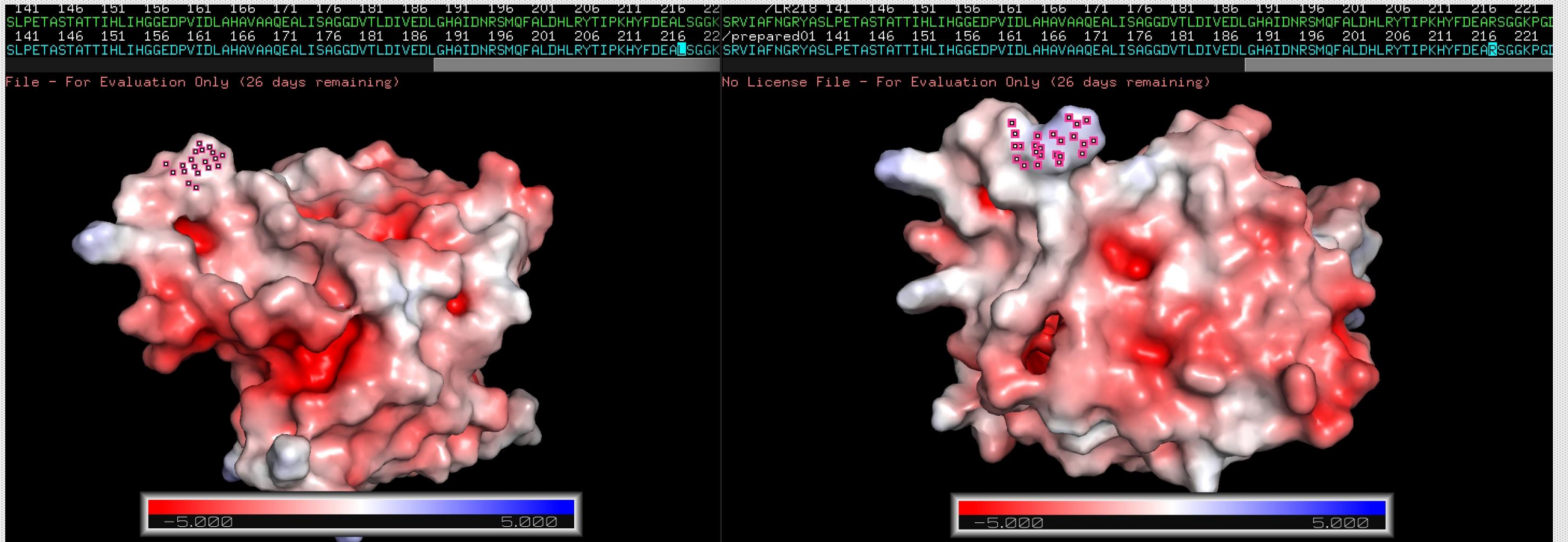
WT



Mutant



LR218 surface potential analysis

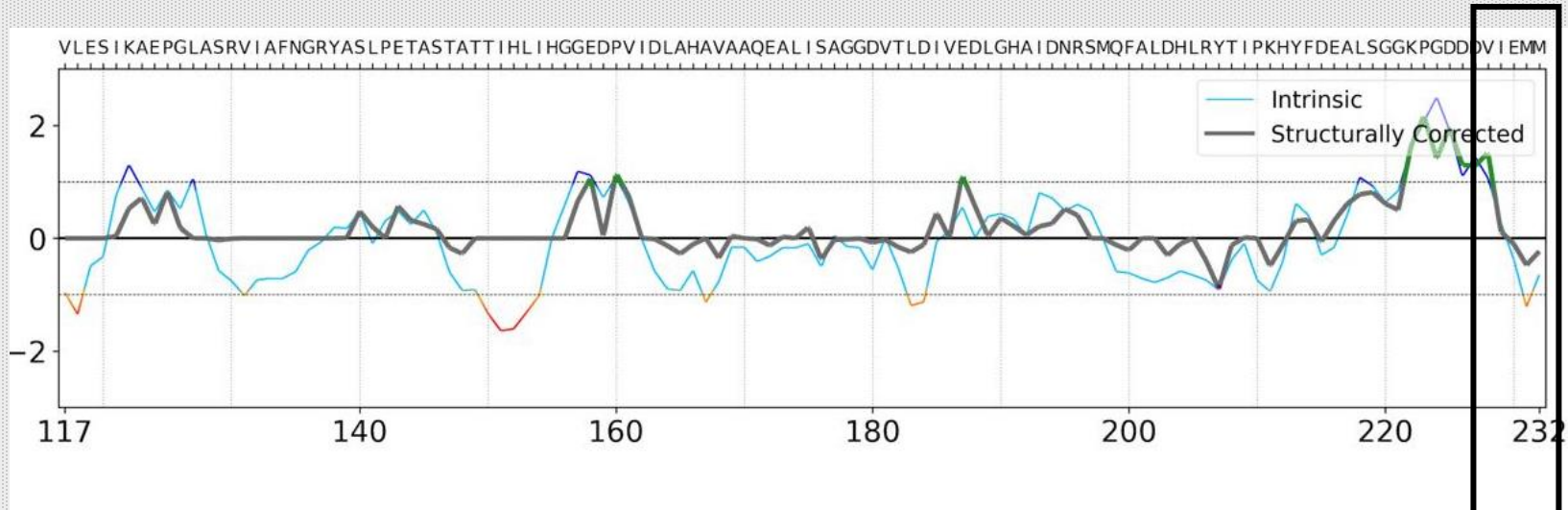


WT

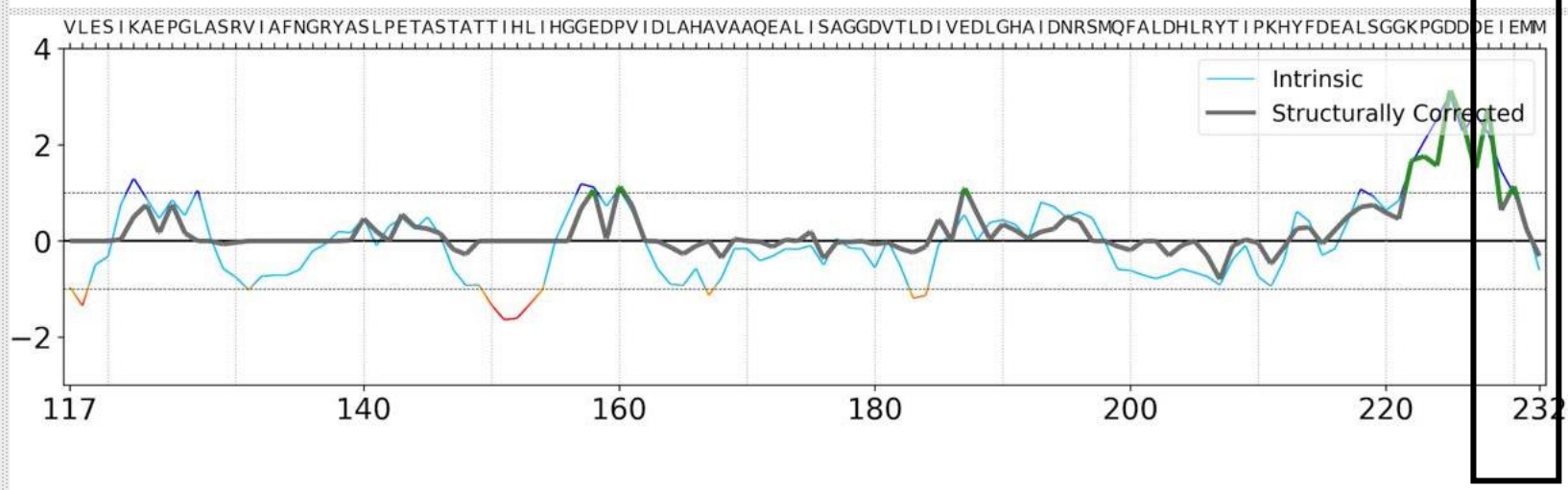
Mutant

VE228 CamSol Profile

WT

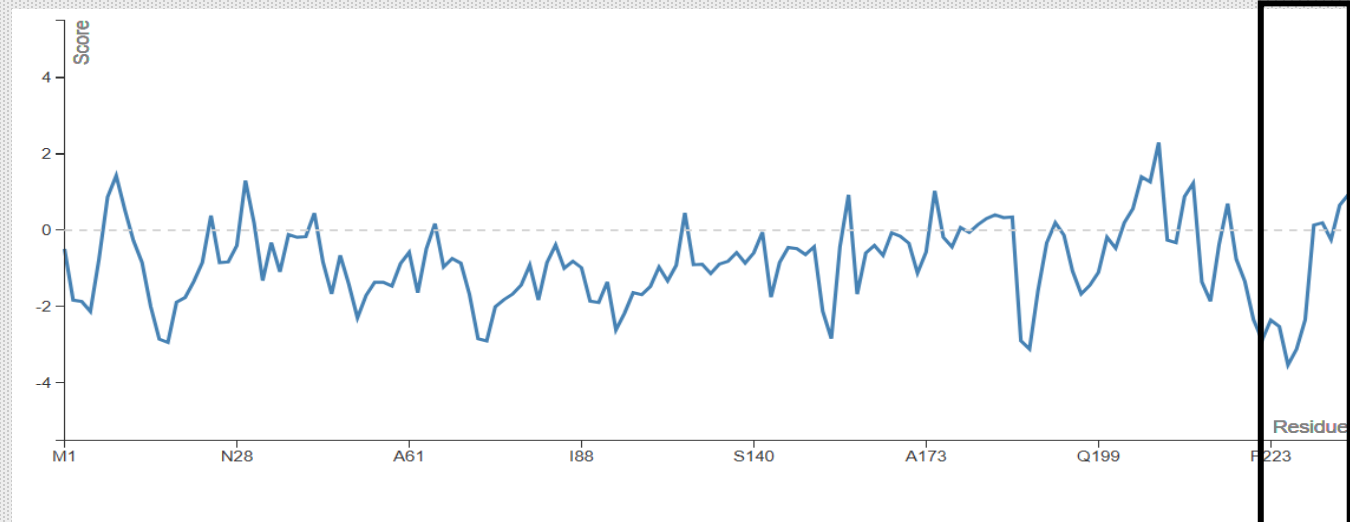


Mutant

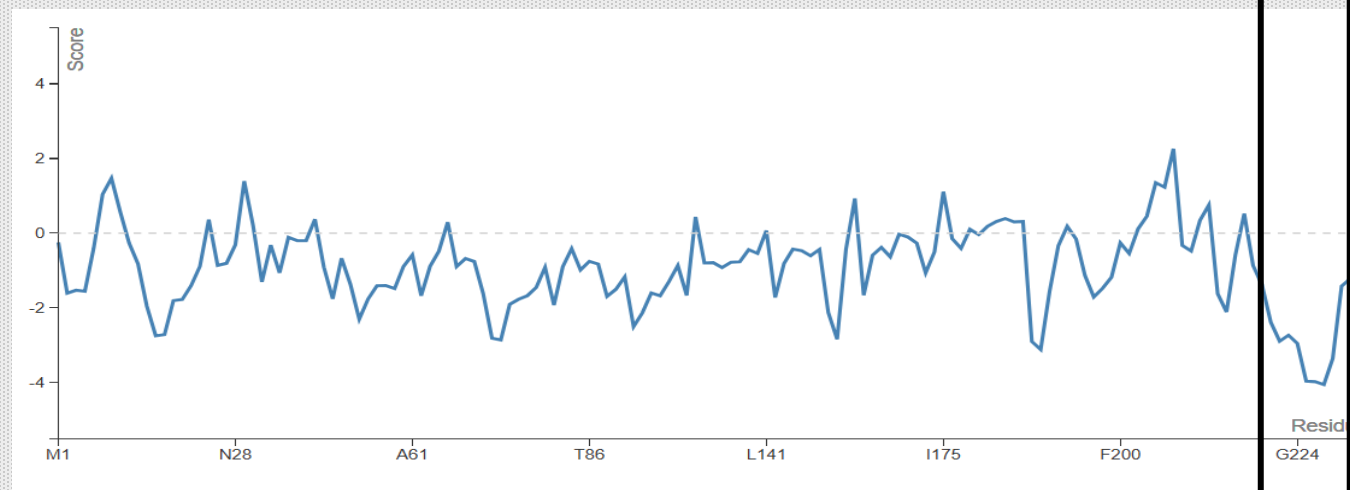


VE228 Aggrescan Profile

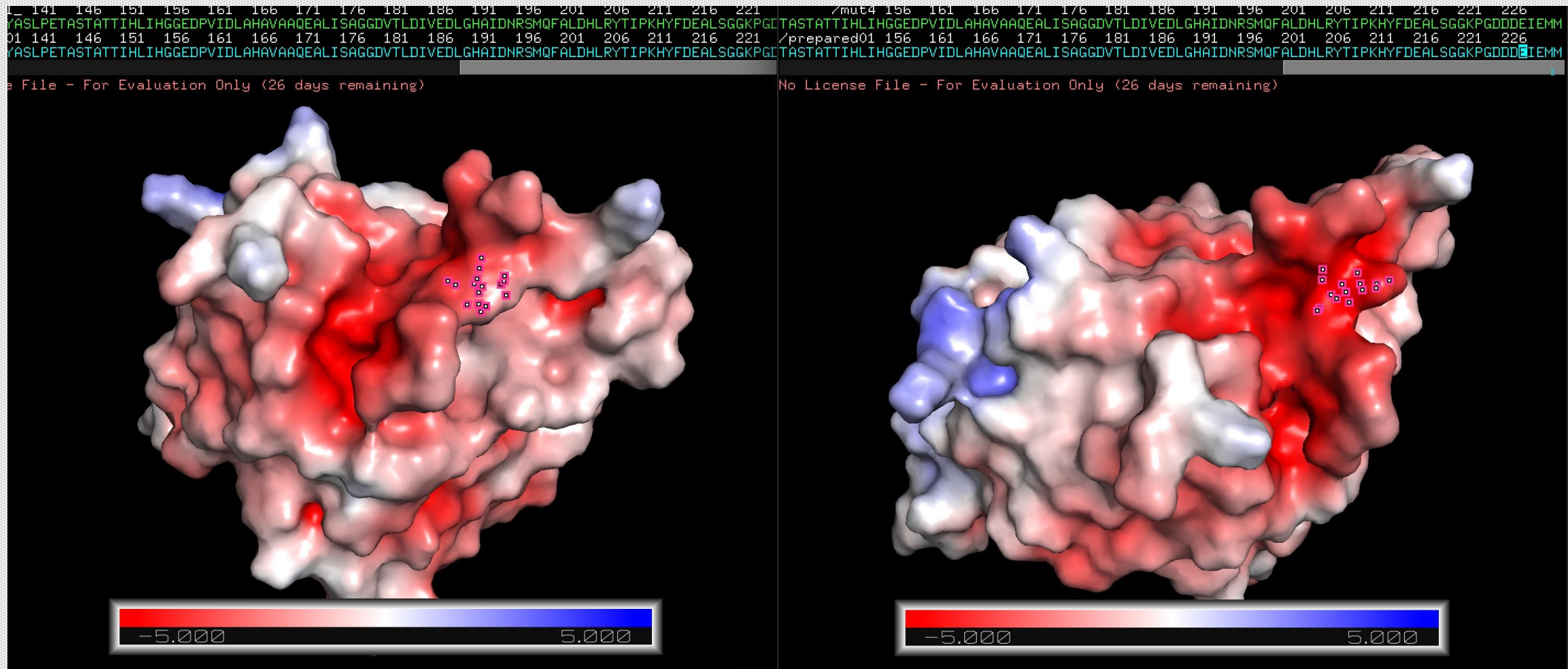
WT



Mutant

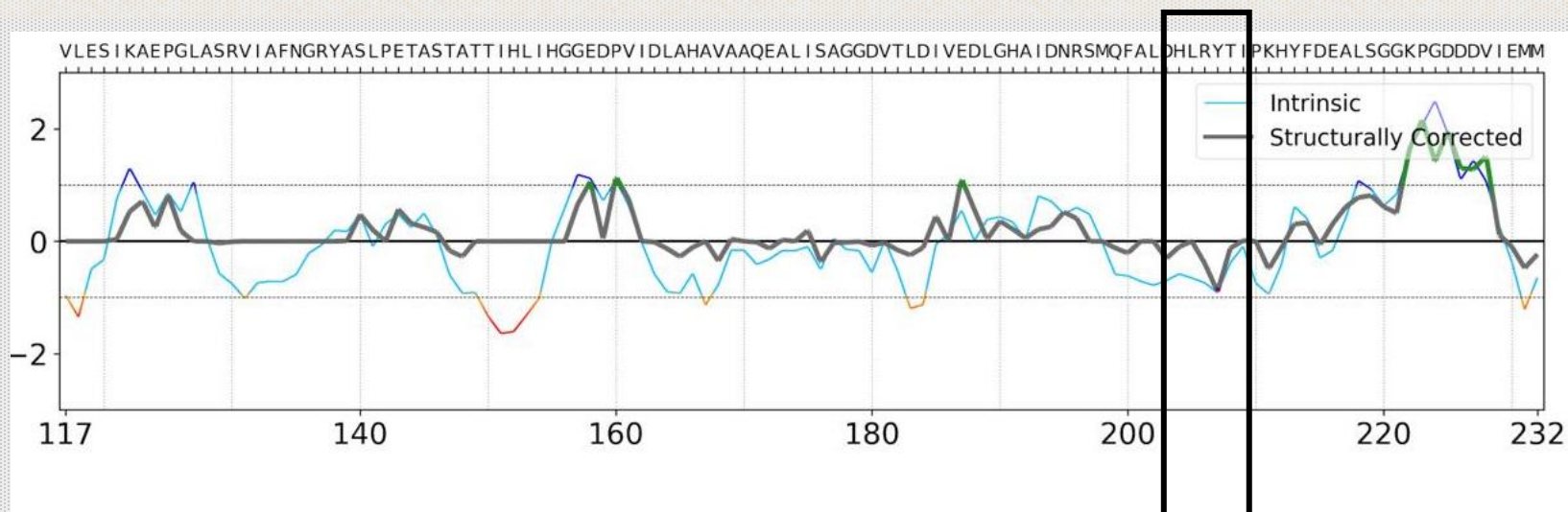


VE228 surface potential analysis

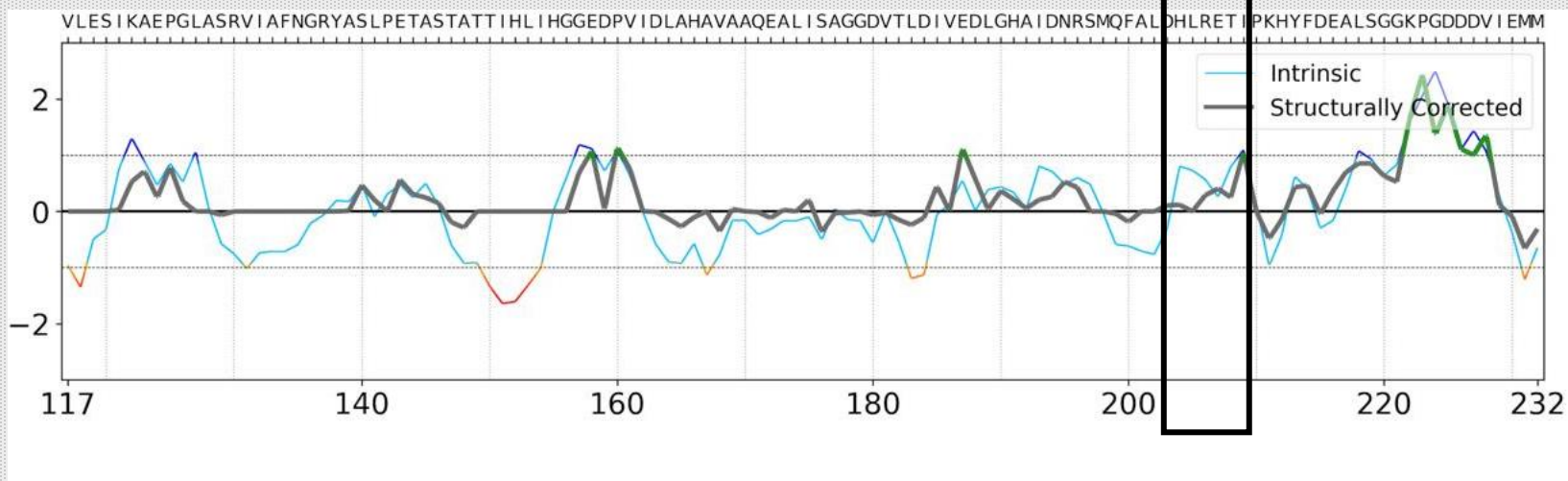


YE207 CamSol Profile

WT

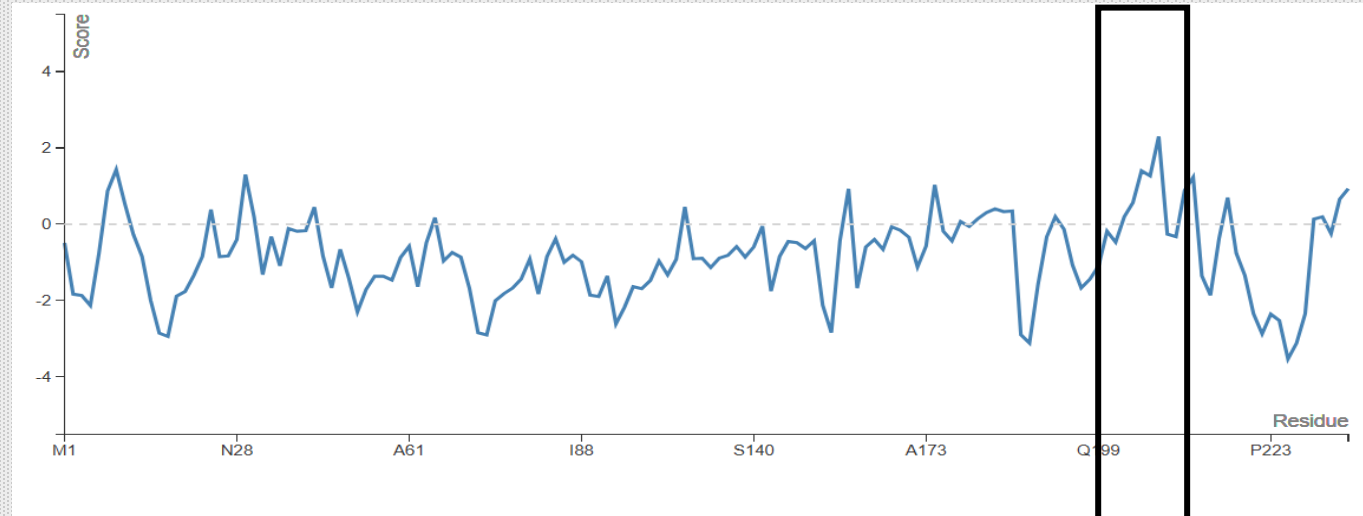


Mutant

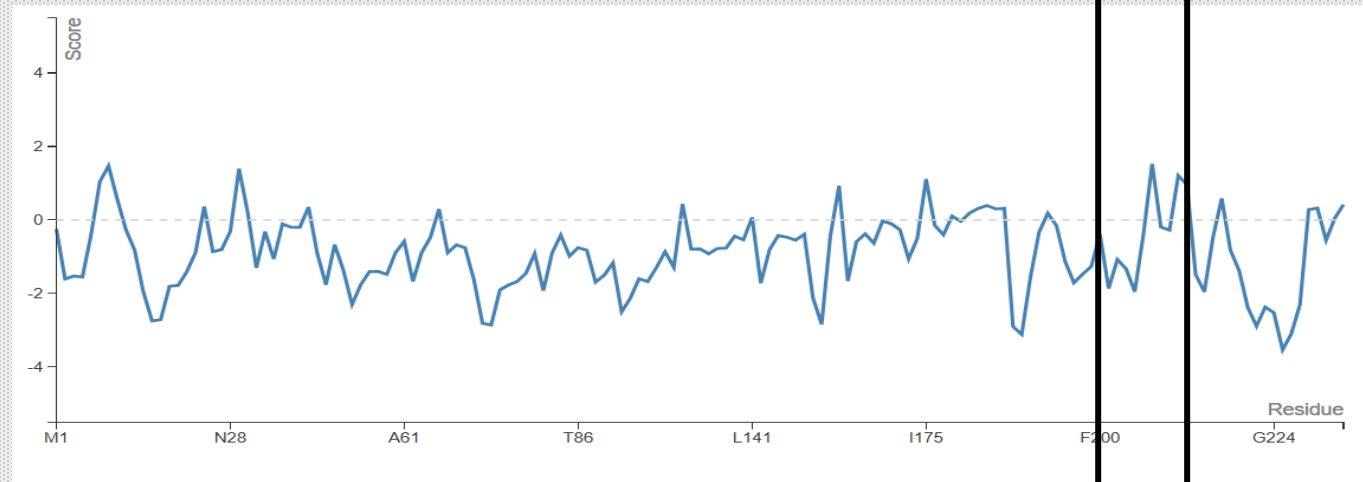


YE207 Aggrescan Profile

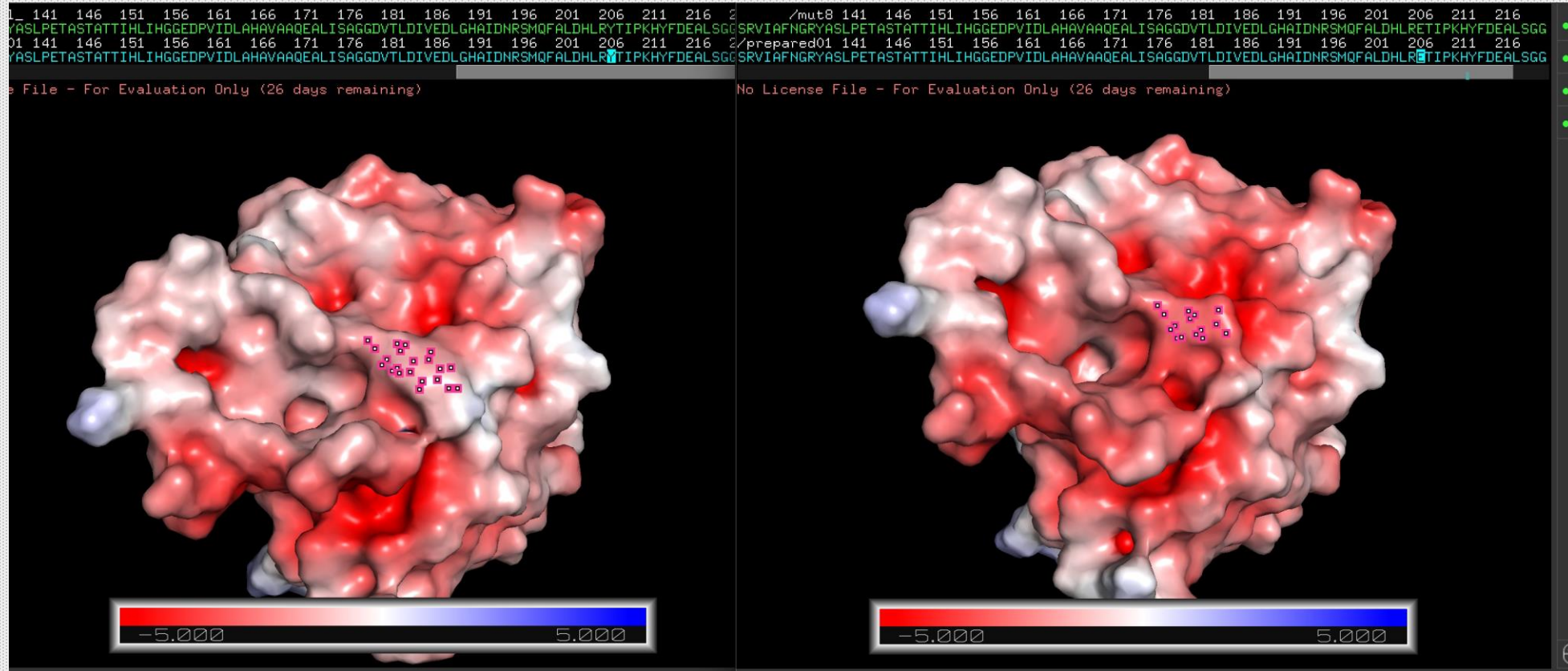
WT



Mutant



YE207 surface potential analysis



WT

Mutant